

HIERARCHICAL GENERALIZED LINEAR MODELS WITH R. APPLICATIONS IN PLANT/ANIMAL BREEDING AND GENETICS

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Recap

- The estimates for an ordinary least squares problem is $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$
- The hat matrix $\mathbf{H}$ “puts the hat” on $\mathbf{y}$, i.e. $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$
- The *hat values* are the diagonal elements of $\mathbf{H}$.
- The estimates for a weighted least squares problem is $\hat{\beta} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}$
- Estimates for a generalized linear model are computed by iterating between weighted least squares

Recap

- Diagnostic plots can be applied on GLMs
- Generalized linear models can be used for a number of distributions
 - Normal
 - Poisson
 - Gamma
 - Binomial
- A Gamma GLM can be used to model the residual variance from a linear model
 - uses *residuals* and *hat values* as input.

Course outline

- **Linear regression**
 - diagnostic plots
- **Weighted least squares**
- **Generalized linear models**
- **Linear mixed models**
 - Estimating fixed and random effects
 - It's a weighted least squares problem!
 - Estimating variance components
 - Application on repeated observations
 - Fitting an animal model using the additive relationship matrix
 - Smoothing

The estrone data

i=1	2	3	4	5
23	25	38	14	46
23	33	38	16	36
22	27	41	15	30
20	27	38	19	29
25	30	38	20	36
22	28	32	22	31
27	24	38	16	30
25	22	42	19	32
22	26	35	17	32
22	30	40	18	31
23	30	41	20	30
23	29	37	18	32
27	29	28	12	25
19	37	36	17	29
23	24	30	15	31
18	28	37	13	32

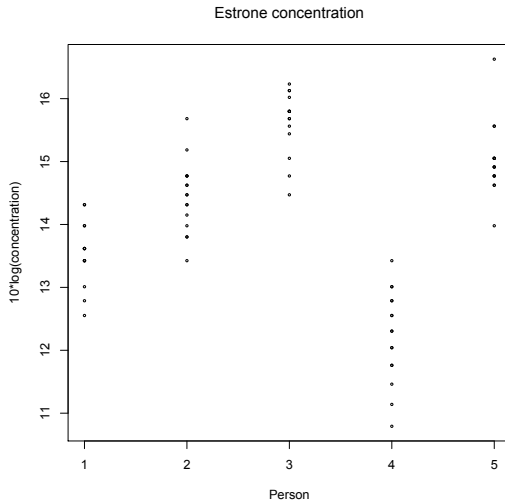
Table 9.1 (Pawitan 2001): Estrone measurements from five menopausal women; there were 16 measurements taken from each woman.

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The estrone data



Model for repeated observations

5 women and 16 observations from each woman.

When we have repeated observations it is natural to model person as random effect

$$y_{ij} = \mu + a_i + e_{ij}$$

y_{ij} = estrone concentration

a_i = person effect, for $i = 1, \dots, 5$

e_{ij} = residual effect, for $j = 1, \dots, 16$

The random person effects are assumed to be iid and normal distributed

$$a_i \sim N(0, \sigma_a^2)$$

The residual are also assumed to be iid and normal distributed

$$e_{ij} \sim N(0, \sigma^2)$$

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Fitting the model in the lme4 package

```
library(lme4)  
  
lmer(y ~ 1 + (1|id), data=estrone)
```

Classical interpretation of fixed and random effects

- A *fixed effect* is fixed and does not change if the experiment is repeated.
- A *random effect* is sampled from a distribution of effects. The effects can change between experiments but the distribution is fixed.

Estrone data example

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Matrix notation

$$y = \mathbf{X}\beta + \mathbf{Z}a + e$$

For the estrone example $\mathbf{X}$ is a matrix having one column of length 80 with all elements equal to 1, β is the same as μ , a is a vector of length 5 including the random effects for each person and $\mathbf{Z}$ is an incidence matrix of size 80x5.

What does $\mathbf{Z}$ look like?

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Incidence matrix for the estrone data example

$$\mathbf{Z} = \begin{pmatrix} \mathbf{1}_{16} & & & & \\ & \mathbf{1}_{16} & & & \\ & & \mathbf{1}_{16} & & \\ & & & \mathbf{1}_{16} & \\ & & & & \mathbf{1}_{16} \end{pmatrix}$$

where $\mathbf{1}_{16}$ is a column vector of length 16 with all elements equal to 1.

Variance matrix of y

$$V(y) = V(\mathbf{Z}a + e) = V(\mathbf{Z}a) + V(e)$$

$$\Rightarrow V(y) = \mathbf{Z}V(a)\mathbf{Z}' + V(e) = \mathbf{Z}\mathbf{Z}'\sigma_a^2 + \mathbf{I}\sigma^2$$

because the random effects and residuals are iid (the estrone data example) with

$$a_i \sim N(0, \sigma_a^2) \Rightarrow V(a) = \mathbf{I}\sigma_a^2$$

$$e_{ij} \sim N(0, \sigma^2) \Rightarrow V(e) = \mathbf{I}\sigma^2$$

How can estimated random effects be used?

- ① Smoothing of time series
- ② Spatial data
 - ① Smoothing
 - ② Predictions for areas with no observations
- ③ Ranking
 - ① “Small Area Estimation” for spatial data
 - ② Health care example: hospital performance for different sized hospitals
 - ③ Genetics:
 - ① performance of dairy cows having different number of relatives
 - ② possible to rank bulls even though they have no observed values on milking performance

Ohio School Example

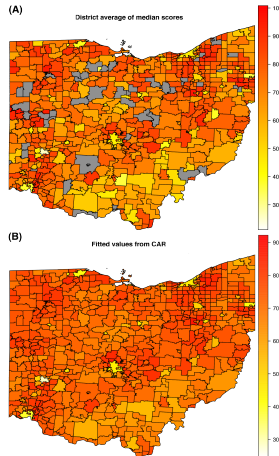
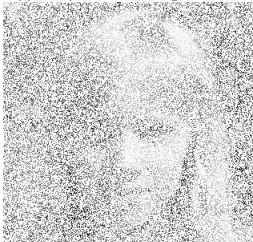


Figure 1: Observed (A) and predicted (B) median 4th grade proficiency scores of the school districts in Ohio. Districts without any observation are displayed in gray.

Connection to image reconstruction

(a) 70 % pixels are missing at random



(b) 4×4 clusters are missing



(c) Reconstruction of (a)



(d) Reconstruction of (b)



Plant Trial Example

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Modelling spatial variation in agricultural field trials with INLA

Maria Lie Selle

Ingelin Steinsland

John M. Hickey

Gregor Gorjanc

April 17, 2019

Key message Established spatial models improve the analysis of agricultural field trials with or without genomic data and can be fitted with the open-source R package INLA.

Plant Trial Example

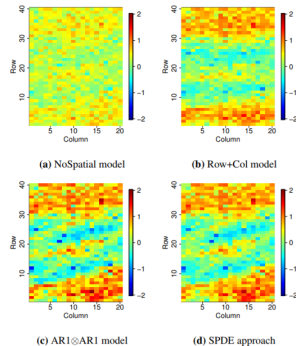
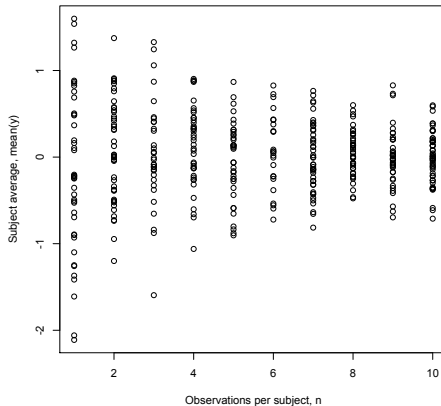


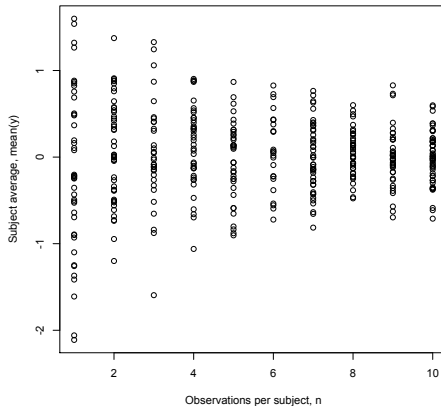
Fig. 5. Posterior fitted values from the model W2 for trial 2011 FI using all three methods of spatial correction and no spatial correction.

Random effects are useful for ranking



A simulated example

Random effects are useful for ranking



Discuss this plot in pairs.

Theory

Let a be a random effect

$y = \mathbf{X}\beta + \mathbf{Z}a + e$ with $a_i \sim N(0, \sigma_a^2)$. Then we have “Henderson’s mixed model equations”:

$$\begin{pmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + I\frac{\sigma_a^2}{\sigma_e^2} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{a} \end{pmatrix} = \begin{pmatrix} \mathbf{X}'y \\ \mathbf{Z}'y \end{pmatrix}$$

Suppose instead that a would be a fixed effect in $y = \mathbf{X}\beta + \mathbf{Z}a + e$. Then the estimates of a are the ordinary least square solutions

$$\begin{pmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{a} \end{pmatrix} = \begin{pmatrix} \mathbf{X}'y \\ \mathbf{Z}'y \end{pmatrix}$$

The only difference is the term $+I\frac{\sigma_a^2}{\sigma_e^2}$ that imposes a shrinkage to the estimates when effect a is random.

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The only difference is the term $+\mathbf{I} \frac{\sigma_a^2}{\sigma_e^2}$ that imposes a shrinkage to the estimates when effect a is random.

Linear mixed model as an augmented regression model

The linear mixed model $y = \mathbf{X}\beta + \mathbf{Z}a + e$ with $a_i \sim N(0, \sigma_a^2)$ can be written as a regression model

$$\begin{pmatrix} y \\ 0 \end{pmatrix} = \begin{pmatrix} \mathbf{X} & \mathbf{Z} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \beta \\ a \end{pmatrix} + \begin{pmatrix} e \\ -a \end{pmatrix}$$

where 0 is a vector of zeros (same length as a), $\mathbf{0}$ is a matrix of zeros and $\mathbf{I}$ is the identity matrix.

The weighted least square solutions are

$$\begin{pmatrix} \mathbf{X} & \mathbf{Z} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}' \mathbf{W} \begin{pmatrix} \mathbf{X} & \mathbf{Z} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{a} \end{pmatrix} = \begin{pmatrix} \mathbf{X} & \mathbf{Z} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}' \mathbf{W} \begin{pmatrix} y \\ 0 \end{pmatrix}$$

where $\mathbf{W} = \begin{pmatrix} \mathbf{I} \frac{1}{\sigma^2} & \\ & \mathbf{I} \frac{1}{\sigma_a^2} \end{pmatrix}$.

Same as Henderson's mixed model equations! We can use the same model diagnostics as for linear regression!!!

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Let

$$y_a = \begin{pmatrix} y \\ 0 \end{pmatrix}$$

and

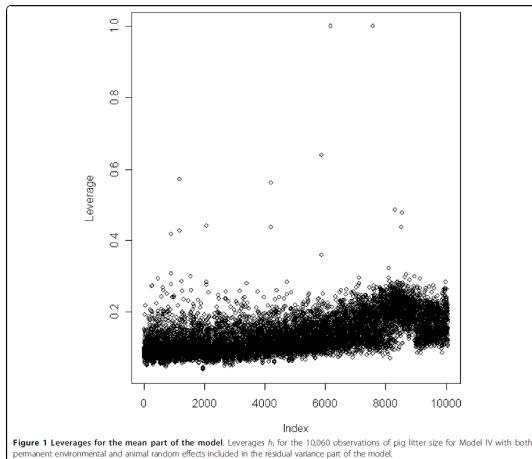
$$X_a = \begin{pmatrix} X & Z \\ 0 & I \end{pmatrix}$$

where the length of y is n and the number of columns in Z is k . Then given the residual variance (`res.var`) and random effects variance (`ranef.var`) the model can be fitted in R using the following code

```
BLUP <- lm(y.a <-.-1,  
data = data.frame(X.a),  
weights = c(rep(1/res.var, n), rep(1/ranef.var, k))  
)
```

Hat values

Rönnegård et al. (2010) GSE 42:8



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Estimating σ^2 and σ_a^2 is just a matter of estimating the residual variance for the augmented linear model. **Rather simple!!!**

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But works only when random effects are not correlated.

We need to iterate because the weights depend on the variance components.

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Fixed effect in a model for the residual variance

The **hglm** package in R computes REML estimates for the residual variance from:

```
glm(d~1, family=Gamma(link=log), weights = (1-h)/2)
```

where $d = \frac{e^2}{1-h}$, h = first n hat values.

We can have explanatory variables for the residual variance!

```
glm(d~x, family=Gamma(link=log), weights = (1-h)/2)
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Notation in the hglm function

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hglm(y, X, Z, X.disp)
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hglm output

```
Call:
hglm.formula(family = Gamma(link = log), fixed = y ~ x1 + x3 +
  x5 + x6, random = ~1 | Device, disp = ~x2 + x3, data = semiconductor)

-----
MEAN MODEL
-----

Summary of the fixed effects estimates:

      Estimate Std. Error t-value Pr(>|t|)
(Intercept) -4.71168    0.06696  -70.368  < 2e-16 ***
x1           0.20979    0.06638   3.160  0.00263 **
x3           0.32893    0.06696   4.913  9.34e-06 ***
x5          -0.17314    0.06638  -2.608  0.01185 *
x6          -0.35690    0.06633  -5.380  1.80e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Note: P-values are based on 52 degrees of freedom

Summary of the random effects estimates:

      Estimate Std. Error
as.factor(Device)1  0.2724  0.1787
as.factor(Device)2  0.0097  0.1787
as.factor(Device)3 -0.2697  0.1584
---
NOTE: to show all the random effects, use print(summary(hglm.object), print.ranef = TRUE).

-----
DISPERSION MODEL
-----

NOTE: h-likelihood estimates through EQL can be biased.

Model estimates for the dispersion term:

Link = Log

Effects:

      Estimate Std. Error
(Intercept) -2.5887    0.1972
x2          -0.6861    0.1971
x3          -0.5024    0.1971

Dispersion = 1 is used in Gamma model on deviances to calculate the standard error(s).

Dispersion parameter for the random effects:
[1] 0.0486

Dispersion model for the random effects:

Link = Log

Effects:
- |Random|
      Estimate Std. Error
-3.0242    0.5172

Dispersion = 1 is used in Gamma model on deviances to calculate the standard error(s).

EQL estimation converged in 4 iterations.
```


hglm output

```

Call:
hglm.formula(family = Gamma(link = log), fixed = y ~ x1 + x3 +
  x5 + x6, random = ~1 | Device, disp = ~x2 + x3, data = semiconductor)

-----
MEAN MODEL
-----

Summary of the fixed effects estimates:

      Estimate Std. Error t-value Pr(>|t|)
(Intercept) -4.71168    0.06696  -70.368  < 2e-16 ***
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```

Variance for
random effects

hglm output

```

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  x5 + x6, random = ~1 | Device, disp = ~x2 + x3, data = semiconductor)

-----
MEAN MODEL
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Summary of the fixed effects estimates:

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```

Random effect
variance on log
scale

Fixed and random effects in a model for the residual variance

We can also include random effects in the model of the residual variance in the **dhglm** package.

See also Rönnegård et al. (2010) *Genetic heterogeneity of residual variance - estimation of variance components using double hierarchical generalized linear models*. GSE 42:8

Some useful R code available at: www.larsronnegard.se

Normal distribution

An observation y_i sampled from a normal distribution $y_i \sim N(\mu, \sigma^2)$ has density function

$$f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2\sigma^2} (y_i - \mu)^2 \right]$$

Normal distribution

A vector of observations $\mathbf{y}$ sampled from a multivariate normal distribution $\mathbf{y} \sim N(\boldsymbol{\mu}, \mathbf{V})$ has density function

$$f(\mathbf{y}) = \frac{1}{|2\pi\mathbf{V}|^{0.5}} \exp \left[-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})' \mathbf{V}^{-1} (\mathbf{y} - \boldsymbol{\mu}) \right]$$

Normal distribution

A vector of observations $\mathbf{y}$ sampled from a multivariate normal distribution $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \mathbf{V})$ has density function

$$f(\mathbf{y}) = \frac{1}{|2\pi\mathbf{V}|^{0.5}} \exp \left[-\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right]$$

Variance-covariance matrix computations

Let $\mathbf{u} \sim N(0, \mathbf{V})$ and $\mathbf{Z}$ is a matrix with known elements.

$$V(\mathbf{u}) = \mathbf{V}$$

The following identity is very useful

$$V(\mathbf{Zu}) = \mathbf{ZVZ}^T$$

Maximizing the marginal likelihood

For the linear mixed model $y = \mathbf{X}\beta + \mathbf{Z}a + e$ the likelihood is

$$\log L(\beta, \sigma^2, \sigma_a^2) = -\frac{1}{2} \log |\mathbf{V}| - \frac{1}{2} (y - \mathbf{X}\beta)' \mathbf{V}^{-1} (y - \mathbf{X}\beta)$$

where $\mathbf{V} = \mathbf{Z}\mathbf{Z}'\sigma_a^2 + \mathbf{I}\sigma^2$.

The variance components, σ_a^2 and σ^2 , are estimated from $\frac{\partial \log L}{\partial \beta} = 0$, $\frac{\partial \log L}{\partial \sigma^2} = 0$ and $\frac{\partial \log L}{\partial \sigma_a^2} = 0$.

Cannot be solved directly and requires an iterative method (eg. Newton-Raphson).

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Derivatives

Differentiation rules for a matrix $\mathbf{M}$ and a scalar x

$$\frac{\partial}{\partial x} \log |\mathbf{M}| = \text{tr}(\mathbf{M}^{-1} \frac{\partial \mathbf{M}}{\partial x})$$

$$\frac{\partial}{\partial x} \mathbf{M}^{-1} = -\mathbf{M}^{-1} \frac{\partial \mathbf{M}}{\partial x} \mathbf{M}^{-1}$$

$$\frac{\partial \log L}{\partial \sigma_a^2} = -\frac{1}{2} \text{tr}(\mathbf{V}^{-1} \frac{\partial \mathbf{V}}{\partial \sigma_a^2}) + \frac{1}{2} (y - \mathbf{X}\beta)' \left(-\mathbf{V}^{-1} \frac{\partial \mathbf{V}}{\partial \sigma_a^2} \mathbf{V}^{-1} \right) (y - \mathbf{X}\beta)$$

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Requires an iterative numerical procedure.

Difficulties: starting values, convergence, $\sigma_a^2 \geq 0$

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